Department of Information Systems
Bachelor of Information Systems
LAB4
IS301 Structured Programming In C

Weighting: 4%
ULO addressed: iii, iv vi, vii
Due date: 11:59 PM Tuesday 13th May 2025
Total Marks: 50

Topics: **Arrays, Search & Sorting & Structures**

Instructions:

1. This is your fourth lab activity.
2. Complete answer the following questions, and submit via Moodle before the due date.
3. You submit a single word document file with the answers and codes in C
4. The questions address topics covered from Arrays, search & sort techniques, and structures.
5. Marks are indicated per question.

Arrays

1. Because of its efficiency, the binary search is the best search for any array, regardless of its size and order. True or False? Explain your answer:

(2 marks)

2. A(n) _____ is an integral value used to access an element in an array. **(1 Mark)**

3. Which of the following array initialization statements is valid? **(1 mark)**

a. `int ary{ } = {1, 2, 3, 4};`
b. `int ary[ ] = [1, 2, 3, 4];`
c. `int ary[ ] = {1, 2, 3, 4};`
d. `int ary{4} = [1, 2, 3, 4];`
e. `int ary[4] = [1, 2, 3, 4];`

Provide a feedback for each of the other options **(2 marks)**

4. Which of the following statements assigns the value stored in **x** to the first element on an array, ary? **(1 mark)**

a. `ary = x;`
b. `ary = x[0];`
c. `ary = x[1];`
d. `ary[0] = x;`
e. `ary[1] = x;`

Provide feedback for each of the other options why you have not chosen. **(2 marks)**

5. Which of the following statements concerning passing array elements is true? **(1 mark)**

a. Arrays cannot be passed to functions, because their structure is too complex.

b. It is not possible to pass just a row of a two-dimensional array to a function.

c. Only the size of the first dimension is needed when a two-dimensional array is declared in a parameter list.

d. When an array is passed to a function, it is always passed by reference (only its address is passed).

e. When a two-dimensional array is passed to a function, the size of the second dimension must be passed as a value parameter.

Provide feed for each option why you haven't chosen them. **(2 marks)**

6. Which of the following statements about two-dimensional arrays is true? **(1 mark)**

   a. A two-dimensional array can be thought of as an array of one-dimensional arrays.

   b. Only the size of the second dimension needs to be declared when the array is used as a parameter.

   c. Two different types can be stored in a two-dimensional array.
   d. The first dimension is known as the column dimension.

   e. When passed to a function, the size of the second dimension must be passed as a value parameter.

   Provide feedback why you choose not to choose the other options. **(2 marks)**

Sorting & Searching

7. The process through which data are arranged according to their values is known as _____. **(1 mark)**

   a. arranging

   b. listing

   c. parsing

   d. searching

   e. sorting

   Provide your feedback why you did not choose the other options **(2 marks)**

8. The _____ sort finds the smallest element from the unsorted sublist and swaps it with the element at the beginning of the unsorted data. **(1 mark)**

   a. bubble

   b. exchange

   c. insertion

   d. quick

   e. selection

Provide feedback for not choosing the other options **(2 marks)**

9. The _____ search locates the target item by starting at the beginning and moving toward the end of the list. **(1 mark)**

   a .ascending

   b. binary

   c. bubble

   d. selection

   e. sequential

Provide feedback for not choosing the other options **(2 marks)**

10. Which of the following statements about a sequential search is false? **(1 mark)**

    a. Any array can be searched using the sequential search.

    b. If the target is not found, every element in the list is tested.

    c. The efficiency of the sequential search is O(n).

d. The list must be ordered.

e. The sequential search is generally recommended only for small lists.

Explain your answer. **(2 mark)**

11. Which of the following statements about two-dimensional arrays is true? **(1 mark)**

a. A two-dimensional array can be thought of as an array of one-dimensional arrays.

b. Only the size of the second dimension needs to be declared when the array is used as a parameter.

c. Two different types can be stored in a two-dimensional array.

d. The first dimension is known as the column dimension.

e. When passed to a function, the size of the second dimension must be passed as a value parameter.

Give an explanation on each of the other options **(2 marks)**

12. What would be printed by the following program? **(1 mark)**

```
#include <stdio.h>

int main (void) {

// Local Declarations
      int list [10] = {0};

// Statements
      for (int i = 0; i < 5; i++)
            list [2*i + 1] = i + 2;
      for (int i = 0; i < 10; i++)
            printf("%d\n", list [i]);
      return 0;
} // main
```

Explain the *for loop* statements **(2 marks)**

13. We have two arrays `A` and `B`, each of 10 integers. Write a function that tests if every element of array `A` is equal to its corresponding element in array `B`. In other words, the function must check if `A[0]` is equal to `B[0]`, `A[1]` is equal to `B[1]`, and so forth. The function is to return true if all elements are equal and false if at least one element is not equal.

(5 marks)

Structures

14. Declare a tagged structure for a student record consisting of five fields: student ID(integer), first name (a dynamically allocated string), last name (a dynamically allocated string), total credits completed (integer), and accumulated grade point average (floating-point).

(2 marks)

15. Declare an array of 12 elements. Each element is a structure with three fields. The first field shows the month in numeric form (1 to 12). The second field shows the name of the month (a dynamically allocated string). The third field shows the number of days in the month. Initialize the array.

(2 marks)

16. Assume we have declared the following structure:

```
typedef struct {
        char x;
        char* y;
        int z[20]; } FUN;
```

Determine which of the following definitions are valid and which are invalid. If invalid, explain the error.

(3 marks)

a. `STRUCT FUN f1;`
b. `STRUCT FUN f5[23];`
c. `FUN f3;`
d. `FUN f4 [20];`

17. Write a function that shuffles a deck of cards. The deck of cards is represented by an array of 52 elements. Each element in the array is a structure for one card, as shown below.

(5 marks)

```
typedef struct {
char* suit; // Clubs, Diamonds, Hearts, Spades
int value; // Ace, 2..9, Jack, Queen, King
} CARD;
typedef CARD DECK [52];
```

The function must use a random number to ensure that each shuffle results in a different card sequence. Hint: Generate a random number in the range 0...51, and then exchange the current card with the card in the random position.